

A readable GSN assurance case

GoalKeeper 0.2.0 · standalone revision and FireSat storyboard

GoalKeeper now gives the assurance argument the primary presentation. The revised baseline replaces the inherited KerML-first exception with GSN v3 core notation, while preserving a requested alternate engineering view.

52

requirements

18

GSN elements

2

source notes

5

open claims

What changed

The SRS and verification procedures were written before implementation. The new design retains the six standard shapes, full claim text and meaningful edge styles. Technical graph and model text are reached through the View menu; the main canvas remains focused on the case.

Area	Result
Primary diagram	Six GSN shapes, full statements, A/J markers and undeveloped diamonds
Relationship rules	Exact GSN Table 1:2-2 matrix; illegal imported edges reported
Screen allocation	Primary GSN canvas; alternate model graph/text in View-menu pop-up
Layout	ELK layered layout; shared symbol dimensions; explicit manual/save/restore workflow
Review outputs	Scalable GSN SVG plus Markdown/JSON and traceable FireSat example

Applicability

This delivery addresses the GSN v3 core profile in Part 1 §2. It does not implement pattern, modular, Assurance Claim Point or dialectic extensions. The FireSat case remains an illustrative draft with operational evidence and regime acceptance outstanding.

Verification status: Automated and observed browser checks passed

Start with the recorded Loss

The example is derived from the prototype Loss Tool, with scope and limits carried into the case.

Loss of authentic fire reports

The resulting case contains 18 GSN elements, 17 core links and two evidence-note references. The source Asset is Fire detection reports (DEMO_AST_FIRE_REPORTS); the local Loss ID is loss, with illustrative HID DEMO_LOSS. The modeled scope is Fire Detection Payload SYS_1.1.2_0, Payload Operating Environment ENV_1.1.2_1, and the Standby and Imaging States.

Source	What it establishes
Loss Tool example createOverview(); commit 8d7641d8e1fe5ef9fcbf7bb912dde4 cdf922fc28	Exact Loss, Asset, modeled nodes/edges and disposition fields
Loss Tool route analysis analyzePaths(); same source commit	Nine terminal routes: four unaddressed RV, two Allowed, two Derived and one Blocked
SSTPA FireSat model 10-space-segment.yaml; HEAD 05e076bfc8 a18e36e146b5ecf01121c32566448c	Payload, environment, States, component, interfaces and function identities
Bundled evidence FS-E1, FS-E2 and loss-source-snapshot.json	Source hashes, inventory, routes, recorded decisions and explicit evidence limits

What the source does not establish

The prototype records 25 nodes and 25 edges. Its nine terminal routes are algorithm results, not independent attacks or probabilities. Simulated UUIDs and illustrative metric inputs are not operational security evidence. The Environment context and tailored-out retired-interface route are excluded by the source route algorithm.

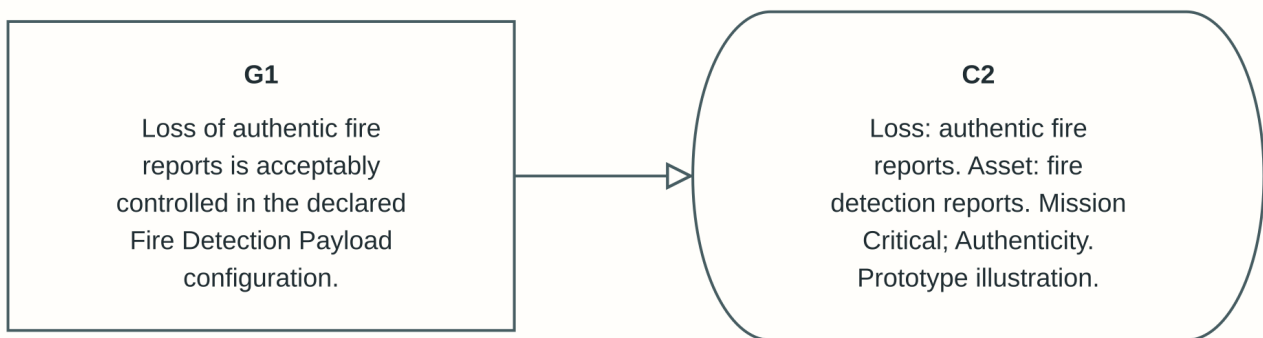
Five claims intentionally remain open

G3 processing, G4 sensor provenance, G5 classification, G6 freshness and G9 authority acceptance retain their undeveloped diamonds. Source notes support traceability and decision-record claims; they do not demonstrate enforcement, control effectiveness or accepted residual exposure.

The six following plates are selected views of the authored case. Omitted nodes are identified in captions; each plate is an authoring/storyboard excerpt, not a declaration that hidden branches are complete.

State the claim

What outcome does the case ask a reviewer to accept?



Selected view of G1 and source Loss context C2. S1 and C1 are omitted from this plate; the top claim remains unproven.

Author action

Begin with the Loss Tool's exact Loss, "Loss of authentic fire reports." Frame a positive assurance claim for the Fire Detection Payload and preserve the source Loss and Asset identities.

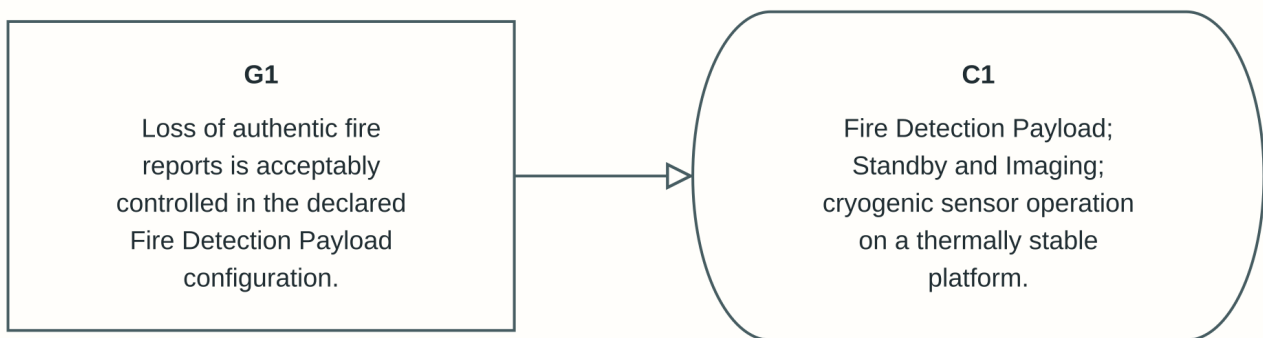
What the reviewer can conclude

G1 is the proposition to be argued. C2 preserves the source Loss. Neither the sentence nor its presence in a GSN rectangle establishes that the claim is true.

Retain the illustrative-draft status; define the scope before developing the argument.

Bound the claim

Which configuration, environment and security property are included?



Selected view of the declared operating scope. Source Loss context C2 and all support branches are omitted; omission does not indicate closure.

Author action

Identify the Fire Detection Payload, its operating environment, and the Standby and Imaging States. Distinguish these operating contexts from an ordered attack scenario.

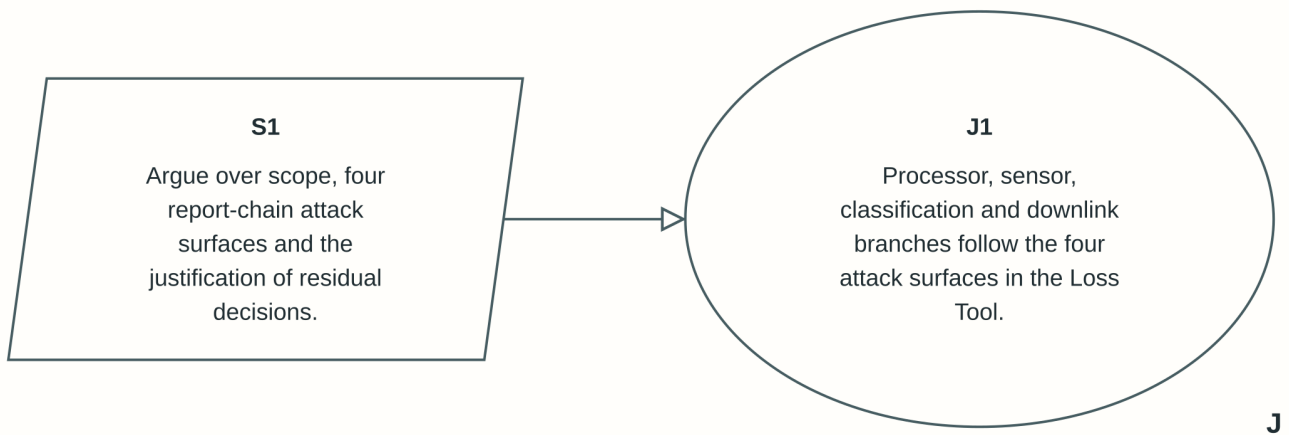
What the reviewer can conclude

C1 constrains interpretation of G1. The recorded payload identities establish the modeled boundary; independent configuration evidence and acceptance criteria remain outstanding.

Keep the source Asset/Loss identifiers and model revision with the case.

Explain the inference

Why does this decomposition address the top claim?



Selected view of strategy S1 and its rationale J1. G1 and the six supporting Goals remain in the full case and are omitted here.

Author action

Use S1 to organize the case around scope, four report-chain attack surfaces and the justification of residual decisions. Record the rationale in J1.

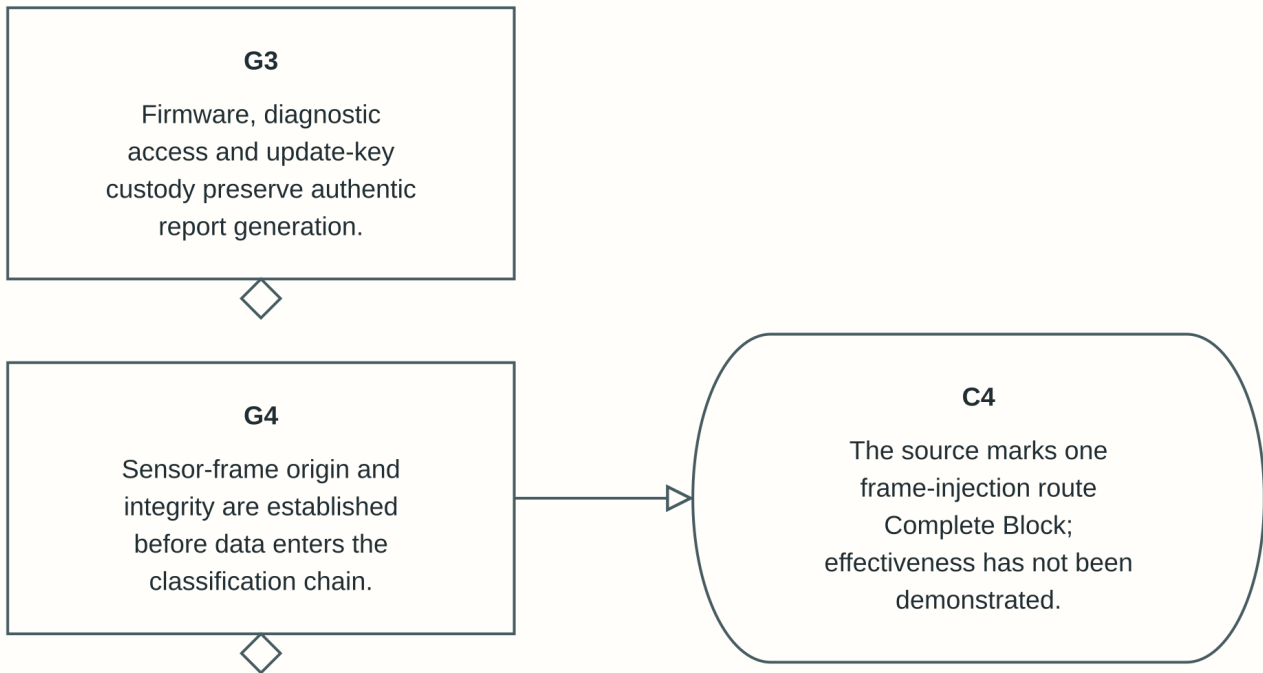
What the reviewer can conclude

The parallelogram carries the inference. The J-marked oval explains the choice of decomposition. Its rationale is bounded to the routes modeled by the source prototype.

Elaborate S1 into supporting Goals; a Strategy cannot directly support a Solution in the GSN core profile.

Expose unfinished branches

Which claims still need operational substantiation?



Selected view of processor and sensor claims. Their undeveloped diamonds remain visible. The classification, downlink and residual-decision branches are omitted here and remain part of the unfinished case.

Author action

Develop processor, sensor, classification and downlink claims. Preserve the undeveloped markers where configuration, enforcement, bypass, replay or key-custody evidence has not been supplied.

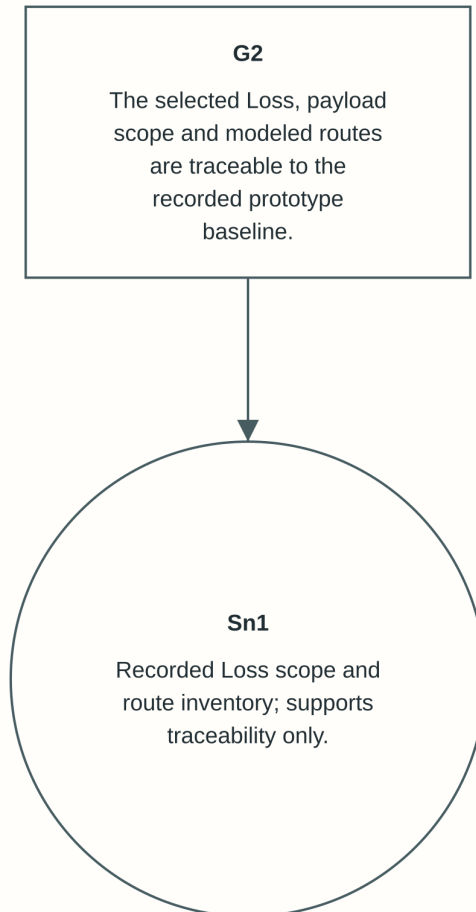
What the reviewer can conclude

G3 and G4 remain open in this excerpt. C4 records that the source marks a frame-injection route Complete Block, while clearly stating that effectiveness has not been demonstrated.

G5 and G6 also remain undeveloped. Named countermeasures alone do not justify removing a diamond.

Attach bounded evidence

Does the cited material establish the claim that relies on it?



Selected view of the traceability claim and source record. FS-E1 supports this narrow record claim; no operational control effectiveness is asserted.

Author action

Attach Sn1 to source record FS-E1 for the scope and modeled-route inventory. Separately, G8 uses Sn2/FS-E2 to document prototype dispositions and rationale.

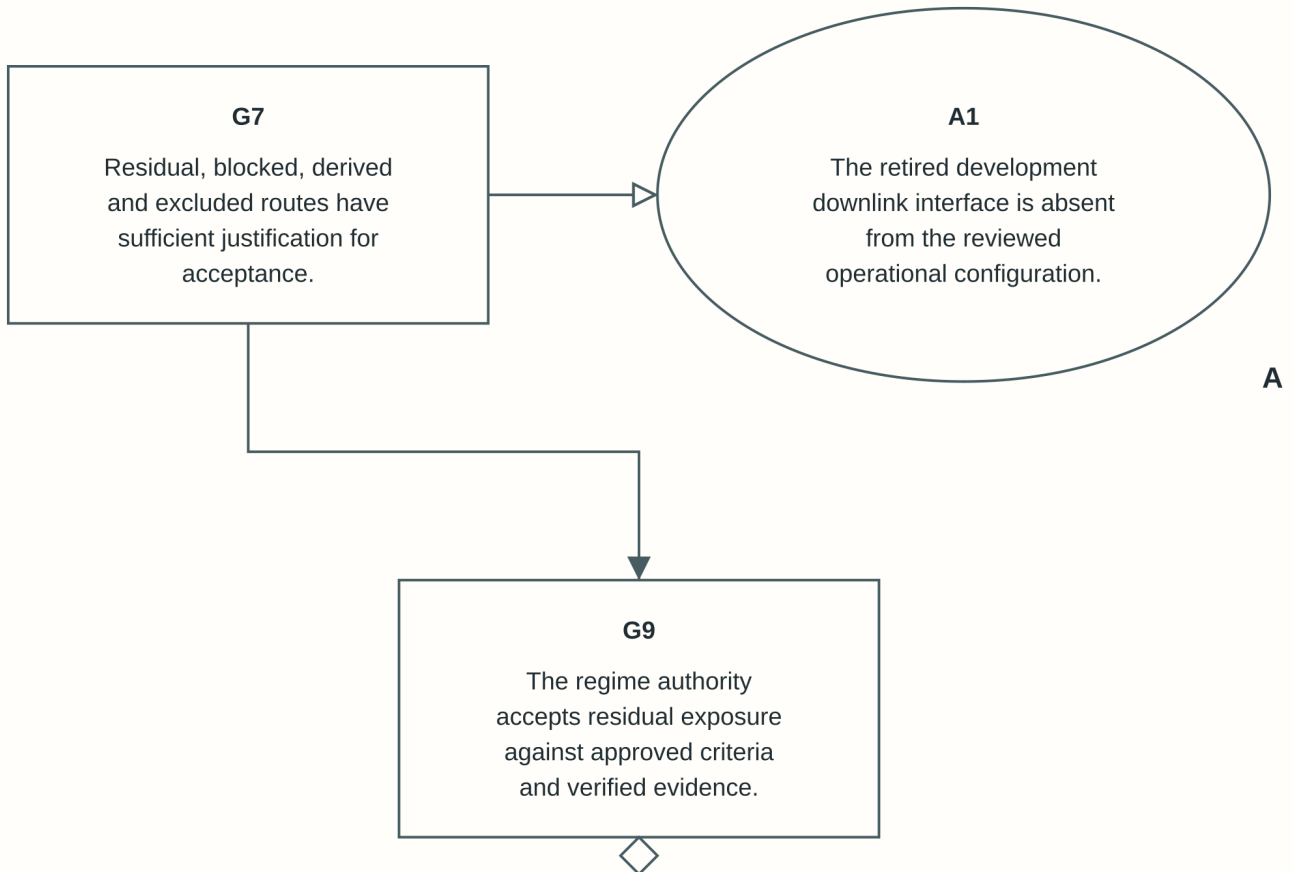
What the reviewer can conclude

The circular Solution references evidence for a narrow traceability claim. Two supplied source notes establish what the prototype records, not that operational controls are effective.

Read the evidence note, artifact path, source revision and limits before relying on any Solution.

Keep acceptance open

What remains unresolved before the case could be accepted?



Selected view of the residual-decision branch. G8/Sn2 and context C3 are omitted. G9 is explicitly undeveloped; the assumption still needs configuration evidence.

Author action

Review residual, blocked, derived and excluded routes. Expose the retired-interface assumption and require a justified independent authority decision against approved criteria.

What the reviewer can conclude

G9 remains explicitly undeveloped. A1 still needs configuration evidence. The recorded disposition facts under G8/Sn2 cannot substitute for acceptance of the residual exposure.

Continue the six-step method recursively as evidence arrives; preserve revision and decision provenance.

Prepared for SSTPA integration

The standalone model and presentation are separated from host identity, storage and authorization.

Boundary	Standalone / future host responsibility
Semantic case	Pure types, rules, graph and Markdown modules. Map local GSN IDs to authoritative Core identities without replacing source provenance
Relationships	Strict support/context matrix. Resolve inherited Strategy→Solution exception before production adoption
Evidence	Local notes and artifact references. Map authoritative evidence to Validation, Verification or Loss records; retain access rules
Persistence	Vault adapter and separate layout metadata. Use the authorized SSTPA transactional commit API and revision handling
Presentation	GSN SVG renderer and ELK layout. Mount in the host tool shell and consume theme tokens; keep model view optional
Model projection	Illustrative read-only graph/text. Use parser-verified G2M/profile output before claiming SysML/KerML interchange

Source conflicts resolved

SSTPA §6.5.11.25's KerML precedence and the old standalone prohibition on GSN shapes are superseded for this application. Strategy→Solution is removed from the allowed support matrix. The inherited docked model panel is replaced by the menu-launched pop-up. The flagship source tree remains a read-only reference.

Distribution and remaining boundaries

ELK.js 0.12.0 uses the same layered engine family as the Loss prototype. Matching source archives and notices ship with an offline checksum gate. Native platform qualification and Rust dependency licensing remain distinct from the JavaScript/ELK package inventory.

SVG, Markdown and JSON outputs preserve the case for review. The alternate model text is illustrative and read-only; a conforming parser and the SSTPA profile are prerequisites for a future language-interchange claim.

Evidence for the revision

GoalKeeper 0.2.0 passed 161 automated tests in 15 files, type checking and production build. Observed browser checks passed at 1440×900. Native compilation passed with 0 tests; native desktop runtime remains unqualified. Tested implementation: 6bc0a0821e74.

Check	Status	Observed result
Automated software checks	PASS	161 tests / 15 files; TypeScript checks; production build. Final npm run verify log retained. Vitest 5.0.3.
Browser GSN geometry	PASS	16 charts, 144 nodes and 1,314 text lines; 0 text outside shapes. Complete text retained at 16px, including long tokens and identifiers.
Observed browser workflows	PASS	At 1440×900, canvas uses 100% workspace width. Pan/drag, saved branch restore, theme reload, pop-up focus, evidence navigation and validation were observed. Keyboard menu launch and Escape focus return passed; 0 browser errors.
Native compilation	PASS	cargo test --offline compiled GoalKeeper 0.2.0 successfully; library, binary and doc harnesses contained 0 tests. This is not native UI qualification.
Dependency/build notices	PASS	13 npm packages; 10 source archives; 11 embedded notices; 9 ELK file hashes; 39 distribution files. npm audit: 0 known vulnerabilities at the recorded run.
Native desktop runtime	PARTIAL	Native picker, physical-vault reopen, application restart and actual desktop interaction were not qualified. Mocked adapter coverage is identified separately.
SVG delivery through browser	PARTIAL	Serializer tests and reopened renderer SVG figures pass. In-app-browser download completion was not observable; no successful download is asserted.
Runtime privacy observation	PARTIAL	Offline/source boundary reviewed. Network observation was unavailable through the browser-control API; no measured no-transmission result is claimed.

What the evidence covers

Automated, browser and native evidence have distinct scopes.

All 52 SRS rows have recorded observations: 42 PASS and 10 PARTIAL, including aggregate parents. Partial rows preserve native/runtime and browser-delivery limits; they are not failed automated tests. See the linked requirement verification matrix and final command logs.

- GSN v3 core Part 1 §2 profile only; pattern, modular, Assurance Claim Point and dialectic extensions are not implemented.
- The alternate SysML/KerML-style projection is illustrative, read-only and not parser-verified language interchange.
- The FireSat argument is an illustrative draft. Operational control effectiveness, accepted residual exposure and authority approval are not established.

SRS and 52 requirement records

Verification procedures

Requirement verification observations

Requirement dispositions

Architecture and integration mapping

FireSat source storyboard

Source snapshot

Dependency notice verification

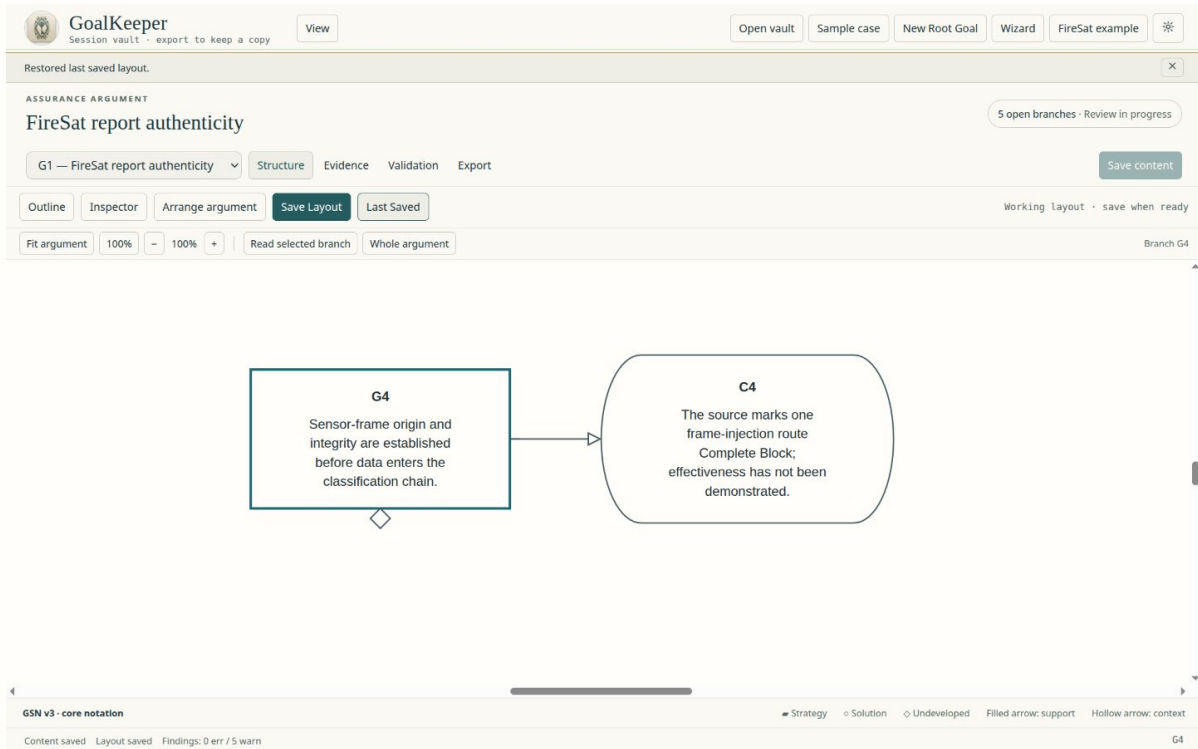
Final software verification log

Native compilation log

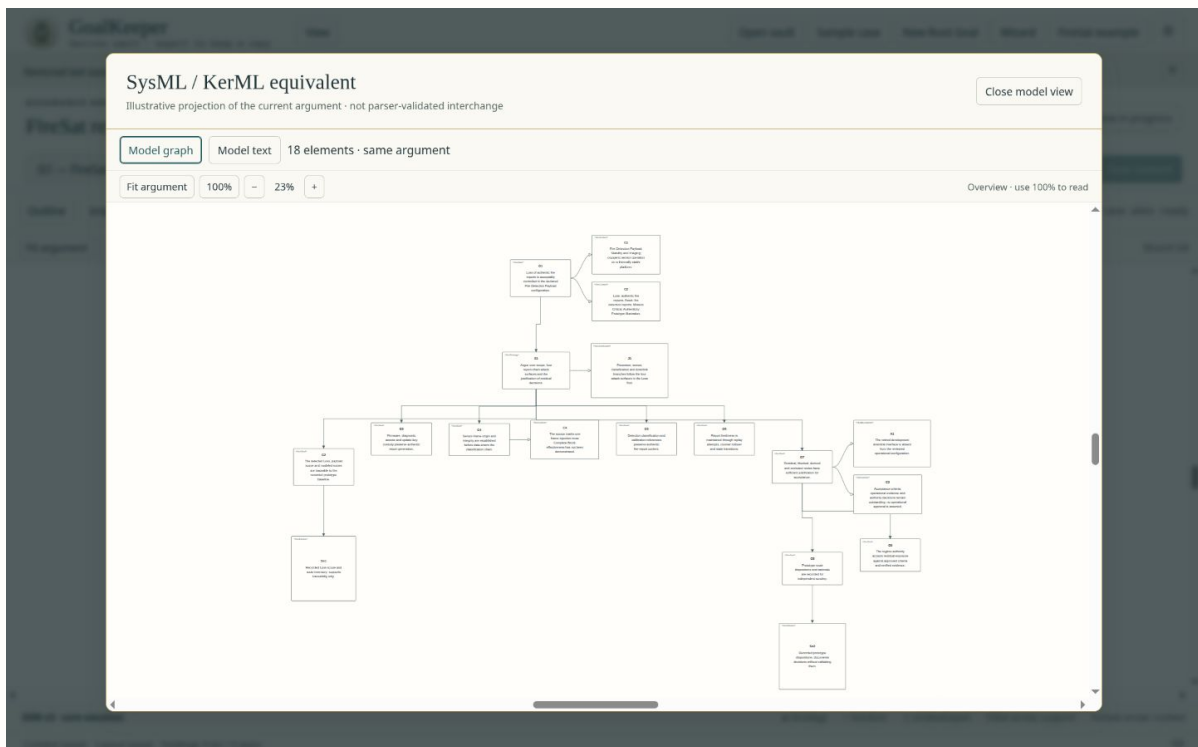
Browser observations and limits

Review in the workspace

Observed application views; the storyboard plates retain the readable claim details.



Primary GSN review - Selected G4 sensor-provenance branch with readable statement, undeveloped marker and source limitation context.



Alternate model on request - The same 18-element argument appears in the View-menu pop-up; model text is explicitly illustrative.

Review in the workspace

Observed application views; the storyboard plates retain the readable claim details.

The screenshot shows the GoalKeeper application interface. At the top, there's a header with the GoalKeeper logo, session vault information, and a 'View' button. Below the header, there's a navigation bar with buttons for 'Open vault', 'Sample case', 'New Root Goal', 'Wizard', and 'FireSat example'. The main content area is titled 'FireSat report authenticity' and shows a 'Solutions & evidence' section. This section contains a 'Loss source inventory' entry with a detailed description of the source and its route analysis. Below the description, there's a JSON artifact representing the source data. The interface also shows a 'Validation' section with various warnings and information messages.

Evidence with its limits - Source record details, provenance and artifact references remain visible to the reviewer.

This screenshot shows the 'Validation' view of the GoalKeeper application. It displays a list of validation findings for the 'FireSat report authenticity' claim. The findings are categorized into 'INFO UNDEVELOPED' and 'WARNING NO SUPPORT'. Each finding includes a description of the issue and a 'Go to' button linking to the relevant goal. The findings are: G3 is explicitly undeveloped; G3 has no supporting node and is marked undeveloped; G4 is explicitly undeveloped; G4 has no supporting node and is marked undeveloped; G5 is explicitly undeveloped; and G5 has no supporting node and is marked undeveloped. The interface also shows a 'Content saved' message and a 'Findings: 0 err / 5 warn' status.

Incomplete work remains visible - 0 structural errors, 5 unsupported-claim warnings and 5 undeveloped information findings.